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Stress Classification Using Heart Rate Variability and Machine Learning Models

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Abstract

Heart rate variability (HRV), representing the time interval between consecutive heartbeats, is a critical biomarker for assessing stress, depression, and anxiety levels. This study makes use of the SWELL-KW dataset, which contains electrocardiogram (ECG) recordings from 25 subjects (17 males, 8 females). ECG recordings were taken in three distinct physiological state: no stress, time pressure, and interruption. HRV parameters were taken of the time, frequency, and nonlinear domains for the classification of stress. The dataset was pre-processed with an interquartile range (25%–75%), which improved resilience and robustness against outliers. The features were adjusted to a 0-1 range using min-max scaling. The hyperparameter tuning was set depending on cross-validation accuracy. Euclidean distances were calculated between class centroids (no stress, time pressure, and interruption), demonstrating distinct differences between the three categories. The present work employed five machine learning models (MLM), namely decision trees (DT), random forests (RF), support vector machines (SVM), K-nearest neighbors (KNN), and adaptive boosting (AdaBoost), to categorize stress levels. The findings of the study shown a remarkable degree of accuracy, surpassing previous projections. Notably, the Random Forest, SVM, and KNN models had exceptional accuracy rates of 1.0, establishing them as the most effective classifiers. On the other hand, the Decision Tree and Adaboost models demonstrated higher levels of accuracy, with values of 0.87 and 0.71, respectively. This study highlights the potential of a proposed methodology on MLM for precise stress prediction using HRV, contributing to advancements in real-time mental health monitoring.

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1. Introduction

Stress, sadness, and anxiety are prevalent concerns within contemporary culture, exerting a substantial influence on both psychological and physiological well-being. Chronic stress activates the sympathetic nervous system, causing physical, psychological, and behavioral problems[1]. Anxiety disorder sufferers typically have a persistently lower

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resting heart rate than healthy individuals. As evidenced by [2] and [3], HRV rises during relaxation and falls under stress. Psychological stress impacts the autonomic nervous system (ANS), which controls the cardiovascular system by means of sympathetic and parasympathetic processes [4]. According to the American Institute of Stress, 80% of employees encounter stress at work on a daily basis and require assistance in coping with it (The American Institute of Stress, 2023). Chronic tension raises the risk of significant health disorders, including diabetes mellitus (DM), dysthymia, cardiac disease, and cancer. [5]. Regularly monitoring an individual's stress level can assist identify high-stress situations and apply early interventions for better stress management. Effective stress management would lower the likelihood of developing serious health problems, hence enhancing the well-being of people in our society. HRV is a non-invasive window into the dynamics of the ANS that is measured by evaluating the time sequence of beat-to-beat intervals from heart rate data. The characteristics of HRV can be strongly impacted by modifications in the neurological system that occur during times of mental stress. Research has demonstrated that stress may be identified using both ultra-short-term HRV analysis less than five minutes and long-term HRV analysis from 24-hour recordings [6], [7], [8]. Numerous studies have looked into the use of HRV as a trustworthy biomarker for stress detection in recent years. [9] collected data from a group of individuals utilizing the Kinect 3D sensor, skin conductance, and ECG. By leveraging the SWELL-KW dataset, they achieved a categorization correctness of 66.52% for KNN, while the SVM with an RBF kernel outperformed it with an accuracy of 72.83%. Using the SVM on the SWELL-KW dataset, [10] obtained an accuracy of 90.0298%. The stress classification methods, which employ various machine learning models, achieve a classification accuracy of 95.75% in distinguishing between no stress, time pressure, and interruption [11]. Currently, no machine learning or deep learning experiments employing the SWELL-KW dataset for multi-group stress categorization have achieved ultra-high validity, notably for stress level partition [12], [13]. Therefore, there is a research void in constructing novel ML models capable of ultra-high-accuracy predictions. This work aims to create and evaluate a robust methodology based on machine learning for the grouping of stress levels using HRV features. Three different situations are used in this work to extract HRV values from ECG recordings: no stress, time pressure, and interruption. The dataset is pre-processed with an interquartile range to remove outliers. Five ML models—DT, RF, SVM, KNN, and Adaboost—are used to classify stress levels from HRV data. These models' accuracy is tested in order to determine the best-performing classifier for real-time stress monitoring. This work is structured as follows: Section 2 provides specifics about the experimental setup, whereas the suggested methodology is described in Section 3 for detecting stress. Part 4 delineates the results, while segment 5 concentrates on the discussion. Conclusions of the paper are found in Section 6. Table 1 summarizes the database used in this study.

2. Experimental Setup

HRV indices were calculated using the SWELL-Knowledge Work (SWELL-KW) [14] to track stress levels in an office setting. Twenty-five people participated in the experiment, doing routine office activities like creating presentations, receiving emails, and writing work reports. Three distinct working conditions: no stress, time pressure, or interruptions. Participants were given unlimited time to complete the required tasks in a stress-free atmosphere. As time pressure increased, the period permitted for work completion was reduced to two-thirds of the time spent in the neutral state. During the interruption period, participants received eight different emails. Certain emails related to assignments that required specific actions, while others were extraneous to the assignment. (Talha Iqbal Physiology Journal). This study utilized the SWELL- KW, compiled by Radboud University. This dataset comprises data from 25 subjects (8 females and 17 males) with a mean age of 25 years having standard deviation is 3.25 years. The experiment registered computer usage habits, facial expressions, and body postures, as well as ECG and electrodermal activity data. Three hours before the recording, the volunteers were told not to smoke or take caffeine. The majority of participants were native of Dutch.

Table 1. HRV Dataset SWELL KW - Summary.

S.No.	Group	Specifics	Number / Summary	Remarks
1	Records	Total number of data records	4,10,322	Collected across all states
2	Features	HRV Features	34	Domains include nonlinear, frequency, and time
3	Domains	Categories of HRV Features	3	Time, Frequency, Nonlinear
4	States	Physiological States	3	No Stress, Time Pressure, Interruption

Nomenclature

SVM Support Vector Machine

KNNK-Nearest Neighbors

IQR Interquartile Range

CV Cross-Validation

3. Proposed Methodology

3.1. Pre-processing of HRV

Prior to analysis, the dataset underwent several pre-processing procedures to guarantee data quality and robustness. The Interquartile Range (IQR) approach was initially employed to identify and eliminate outliers [15]. The IQR filtering method preserved data between the 25th and 75th percentiles, minimizing the impact of outliers that could distort the study. Min-Max Scaling was employed on the HRV features to standardize the dataset, normalizing all values to the 0-1 range. This phase was essential to guarantee that all features contributed uniformly during the model training process, preventing any single feature from unduly affecting the outcomes due to variations in scale. The Kaggle HRV dataset is used in the research [16]. The data processing phases resulted in 25 participant data, consisting of 410322 rows and 34 attributes designated for stress level categorization. These datasets were used to calculate HRV parameters as described by Nkurikiyeyezu et al. [16]. Thirty-four HRV features from frequency, time, and nonlinear were used to classify stresses. These characteristics provide thorough insights into the autonomic nervous system (ANS) reactions of the subjects under varying stress levels.

3.2. Model training and testing

The three stress levels—no stress, denoted as 0, time pressure, denoted as 1, and interruptions, denoted as 2—were distinguished using the HRV attributes. The final column of the prepared data set contained the target labels for these classes. The HRV indices from all three domains were included in the dataset. For the aim of stress categorization, five models were chosen: DT, AdaBoost, KNN, RF, and SVM Classifier. Each model was assessed based on its ability to accurately identify the given three physiological states using the HRV feature dataset. The Adaboost technique's fundamental concept is to train several weak classifiers on the same dataset, then Subsequently, integrate these feeble classifiers to construct a robust classifier.[17]. A supervised learning algorithm is called DT. It can reduce a complicated decision-making process to a set of straightforward choices. This results in an interpretable solution [18]. Diversity in the Random Forest is achieved by randomly selecting training data for each tree and by independently choosing subsets of variables at each node of the tree. The display of individual trees is achieved by employing unpruned tree classifiers with the greatest splitting depth [19]. SVM is a powerful labeling technique that finds the feature space hyperplane that best separates the type. KNN is a simple, instance-based learning technique that assigns the group label of a test sample based on the predominant category of its nearest neighbours, utilizing a distance metric such as Euclidean distance. The dataset was partitioned into training and testing sets with an 80:20 fractional split ratio. Grid search cross validation (CV) is used to perform hyperparameter tuning on each model in order to determine the optimal configuration. Each model is tested using a 5-fold cross-validation process on the training set. The best hyperparameters are then identified, the model is evaluated using the test set, and a variety of performance measures are computed. For every model, the confusion matrix is plotted with unique labels that correspond to all categories, such as No Stress, Time Pressure, and Interruption. Hyperparameter tuning was conducted using Grid Search CV, a systematic approach that evaluates various combinations of hyperparameters to identify the best configurations for each model. The hyperparameters in the Decision Tree algorithm determine the maximum depth of the tree. The random forest's tree count, the SVM's regularization parameter (C), kernel type, and kernel coefficient (gamma), the KNN's number of neighbors, and the AdaBoost algorithm's number of estimators and learning rate. During the grid search, 5-fold CV was taken to make sure the models could generalize well to new data and were not overfitting. Each

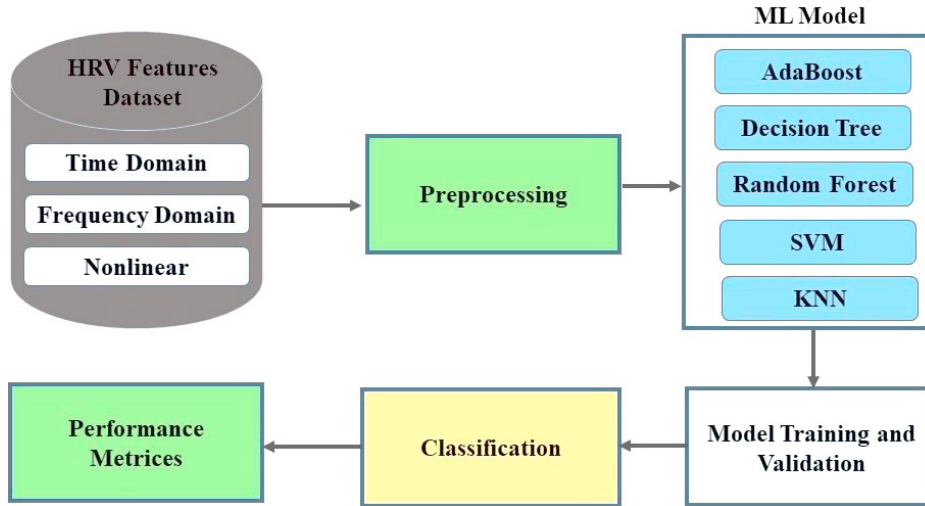


Fig. 1. Block diagram of proposed methodology.

model underwent evaluation using the testing set, and performance measures such as accuracy, precision, recall, and F1-score were assessed. Analogue confusion matrices were created to visually represent the categorization performance at the three different stress levels. Furthermore, the cross-validation scores were calculated to evaluate the level of consistency of the models across several folds. The metric employed to assess the proximity or divergence between distinct classes is referred to as clustering. The Euclidean distance between the centroids of the feature distributions for No Stress, Time pressure, and Interruption is calculated. The Euclidean distances among the various ECG feature vectors are computed as outlined in (1).

$$x(a, b) = \sqrt{\sum_{i=1}^m ((u_i - v_i)^2)} \quad (1)$$

Let x denote the Euclidean distance, u and v represent the arrays, and m signify the array dimension.

This work assessed the performance of multi-class stress categorization using discrimination analysis on the SWELL-KW HRV dataset. Discernment analysis measures include accuracy (Equation 2), precision (Equation 3), recall (Equation 4), F1-score (Equation 5), and confusion matrix. A confusion matrix is a tabular representation of 2-dimensional data, comparing actual values with anticipated values. Both dimensions of the matrix include four possibilities [20]: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1 Score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \quad (5)$$

4. Result

The HRV dataset, which consists of 34 features, both linear and nonlinear, was used for this study[16]. Proposed methodology accuracy is illustrated in Table 2. The study results, which can be seen in the confusion matrices (Figs. 2,

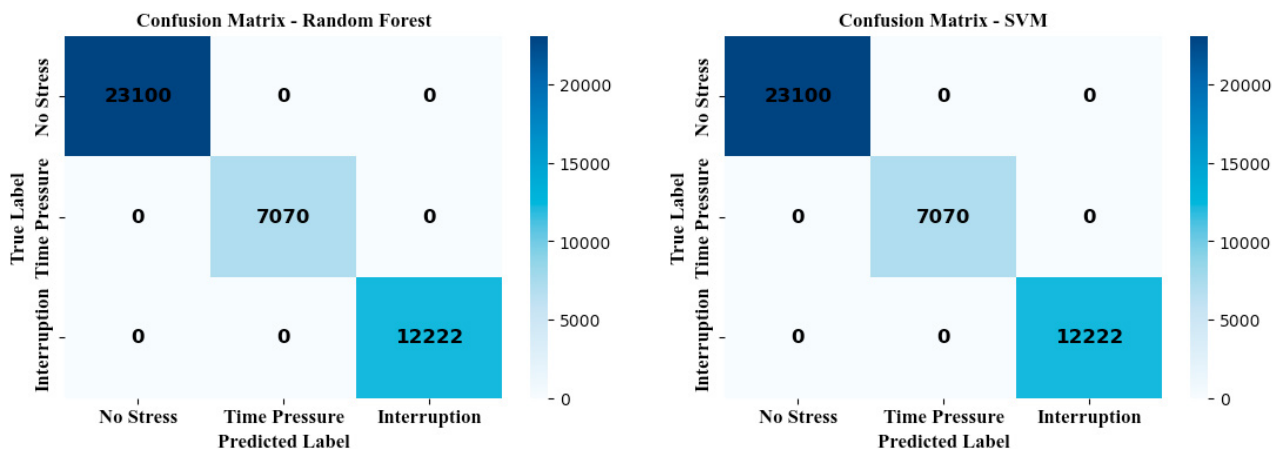


Fig. 2. (a) Random Forest - Confusion Matrix (b) SVM - Confusion Matrix

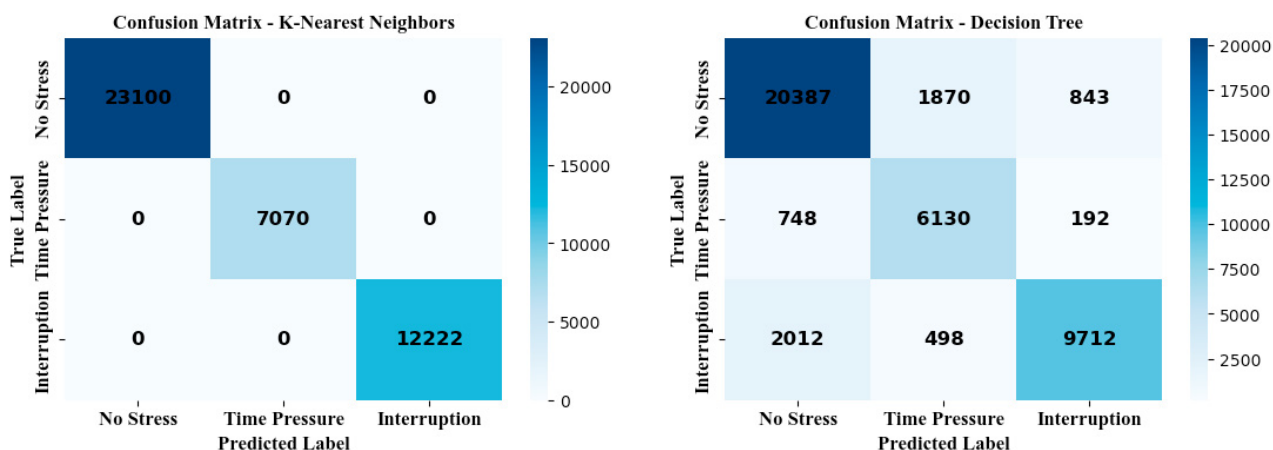


Fig. 3. (a) KNN - Confusion Matrix (b) Decision Tree - Confusion Matrix

3 and 4a) and performance metric plots (Figs. 4b to 6), give us important information about how well different machine learning models can use heart rate variability measurements to accurately detect stress. The uniformity and precision seen throughout the models highlight the reliability of the chosen machine learning model and the possibility of real-time implementation in stress monitoring systems. The confusion matrix of the AdaBoost model presented in Fig. 4a shows a cross-validation accuracy of 0.71 and a testing accuracy of 0.72, suggesting a moderate level of performance. The Decision Tree model in Fig. 3b has a better average cross-validation accuracy of 0.87 and a testing accuracy of 0.85, which can be seen by how well it works across all folds. The RF model (Fig. 2a) attains approximately flawless performance, with an average cross-validation accuracy of 0.999 and a testing accuracy of 1.0. Furthermore, the SVM model (Fig. 2b) and the KNN model (Fig. 3a) demonstrate exceptional performance, with an average cross-validation accuracy of 0.999 and a testing accuracy of 1.0. The results illustrate the exceptional classification abilities of the Random Forest, SVM, and KNN models in this work. It is evident from Fig. 7's interclass distances that No Stress is distinct from Time Pressure and Interruptions classes, with the distances between No Stress and Time Pressure being 1545.03, Time Pressure and Interruption being 1027.33, and Interruption and No Stress being 688.59.

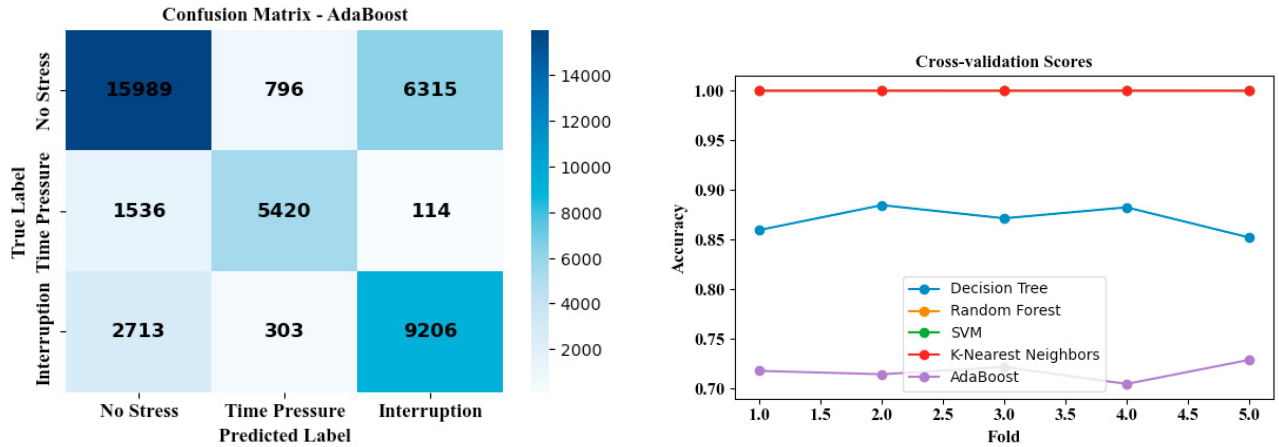


Fig. 4. (a) AdaBoost - Confusion Matrix (b) Accuracy vs. Five-Fold Cross-Validation

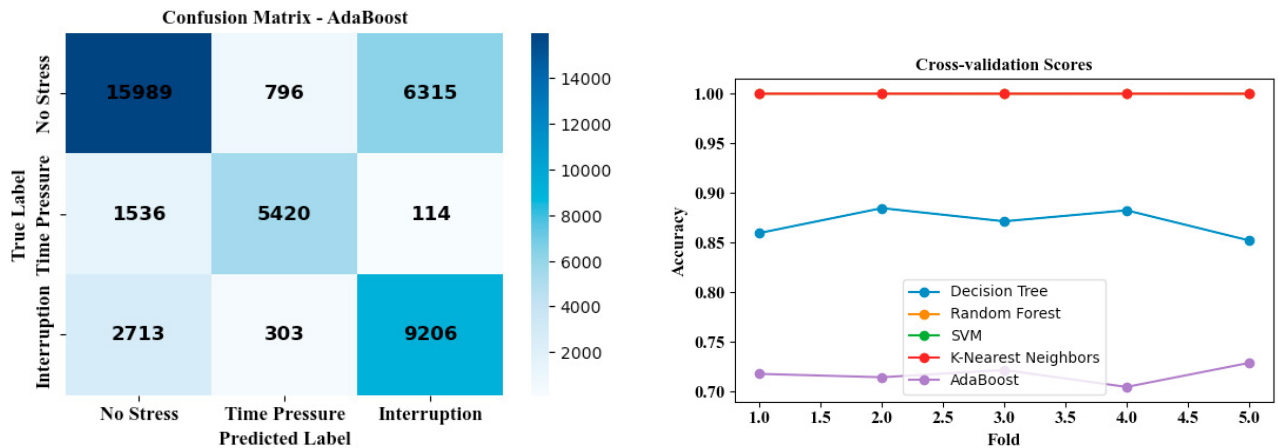


Fig. 5. (a) F1-Score vs Five-Fold Cross-Validation (b) Precision vs Five-Fold Cross-Validation

Table 2. Performance of Machine Learning Model.

Sl.No	Mode	Accuracy	Precision	Recall	F1 Score
1	AdaBoost	0.72	0.74	0.72	0.73
2	Decision Tree	0.85	0.86	0.85	0.86
3	Random Forest	1.0	1.0	1.0	1.0
4	Support Vector Machine	1.0	1.0	1.0	1.0
5	K-Nearest Neighbors	1.0	1.0	1.0	1.0

5. Discussion

This study aimed to assess how well machine learning algorithms classified stress, time pressure, and interruption levels based on HRV features. Ensuring that models generalize effectively to unknown data is a crucial part of machine learning advancement, and this was the main objective of the current study. Establishing acceptable degrees of generalizability across various machine learning methods is just as crucial as achieving high generalizability. With the SWELL-KW dataset, the average cross validation of 5-fold is 0.999, and the testing accuracy is 1.0 for these three models: Random Forest, SVM, and KNN. These models got perfect testing accuracy and showed excellent generalizability. In contrast, AdaBoost and Decision Tree models showed only moderate performance, indicating the variation

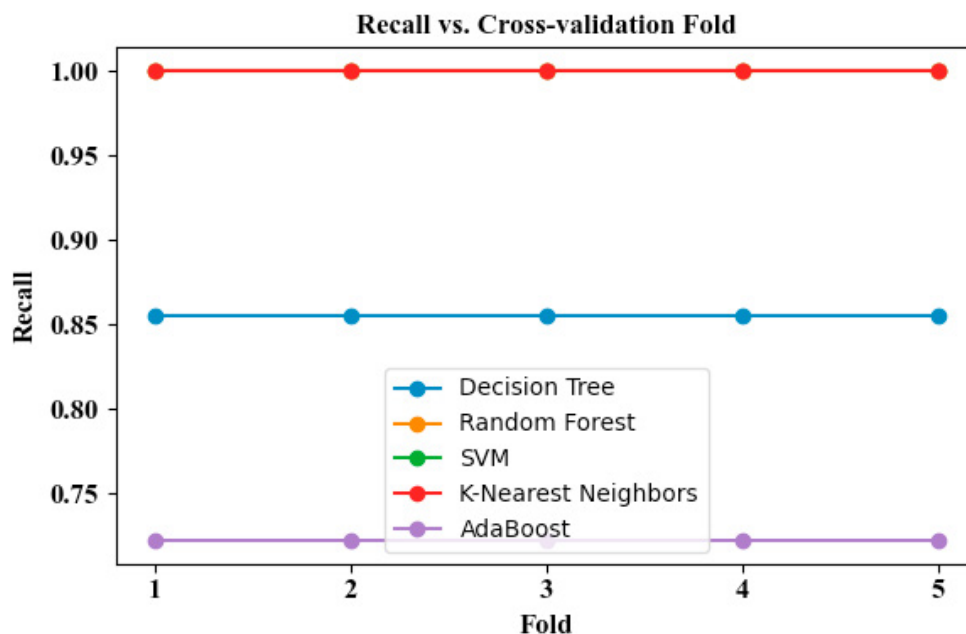


Fig. 6. Recall vs Five-Fold Cross - Validation.

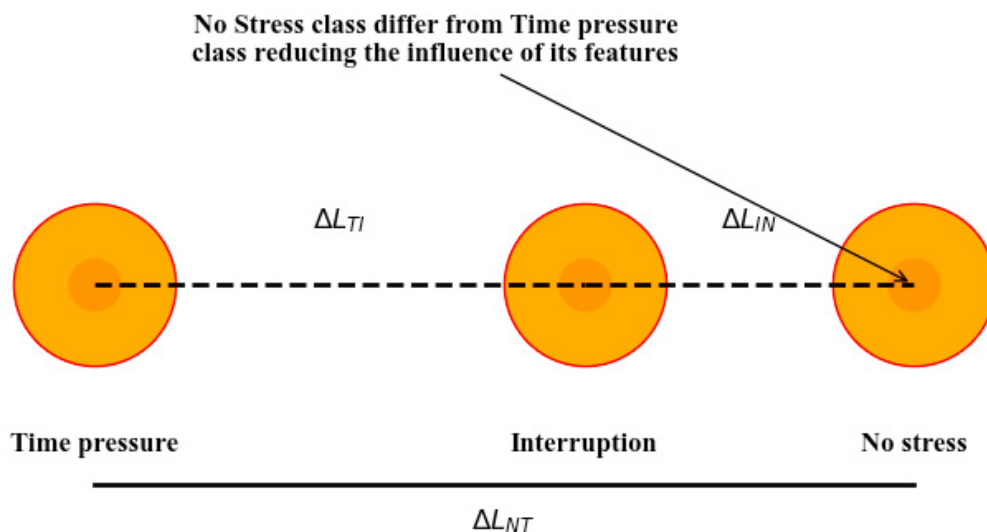


Fig. 7. Interclass distance

in how different algorithms adjust to the intricacies of HRV data. These results highlight the necessity of selecting and fine-tuning machine learning models carefully in practical applications, where the ability to generalize to new data is essential. Based on the cross-validation findings of the different machine learning models, it is clear that the performance differed greatly among the models. This variation indicates their capacity to adapt and provide appropriate handling of the HRV data from the SWELL-KW dataset. The AdaBoost model exhibited a reasonable level of generalizability, as evidenced by five-fold cross-validation accuracies dropping between 0.7049 and 0.7291. As a result, we estimated the average accuracy to be approximately 0.718. These findings suggest that although AdaBoost had some capacity to represent the data, it faced difficulties in maintaining consistency over many folds. Conversely,

the Decision Tree model demonstrated improved performance, with accuracies ranging from 0.8523 to 0.8848 and an average accuracy of 0.8702, indicating superior generalization and stability characteristics. Significantly, the Random Forest, SVM, and KNN models had outstanding performance, achieving almost flawless accuracies in CV. The Random Forest model attained an average accuracy of 0.99999, the SVM model obtained 0.99998, and the KNN model achieved 0.99997. These results demonstrate the models' robustness and high reliability in identifying stress levels. The findings indicate that ensemble techniques such as Random Forest and non-linear classifiers like SVM and KNN are highly suitable for this classification task. They consistently and accurately predict outcomes across various data subsets, making them well-suited for practical use in stress detection. Drawing from the SWELL-KW HRV dataset, Table 3 provides a comparative analysis of prior studies.

Table 3. Summary Table Title Here

Sl.No.	Author(s)	ML Model(s) Used	Features Used	Accuracy (%)	Dataset
1	Sriramprakash et al.[9]	KNN, SVM	HRV Features	72.82	SWELL- KW
2	Talha et al.[20]	KNN	HRV Features	75	SWELL - KW
3	Mariano et al. [12]	MLP	HRV Features	88.64	SWELL - KW
4	Soharabanu et al.[21]	KNN	HRV Features	99	SWELL - KW
5	Mortensen et al.[22]	1D-CNN	HRV Features	99.99	SWELL - KW
6	Proposed Method	SVM, KNN, Random Forest	HRV Features	100	SWELL -KW

6. Conclusion

Accurate and resilient stress recognition models are required due to the health benefits of early stress detection and mitigation. This study designed and assessed various machine learning models for categorizing stress levels—no stress, time pressure, and interruption—using HRV attributes derived from the SWELL-KW dataset. The methods utilized multiple models, including Random Forest, SVM, KNN, Decision Tree, and AdaBoost, to guarantee accurate and dependable stress detection. This research devised and assessed various machine learning models for classifying stress levels. An exhaustive assessment was conducted utilizing criteria like accuracy, recall, precision, and F1 score to evaluate the models' efficacy in differentiating between stress levels. The outcomes of the experiment showed that the Random Forest, SVM, and KNN models performed almost flawlessly, with metrics for the multi-class classification task approaching 100%. These results highlight the accuracy and dependability of the suggested method for stress detection. Cross-validation was another method used to verify the generalizability of the models, and it was shown that Random Forest, SVM, and KNN consistently performed better than other models in both the training and testing stages. The research holds relevance as it demonstrates how HRV features, in conjunction with sophisticated machine learning approaches, can provide a non-invasive and efficient alternative for stress monitoring. This work lays a solid platform for real-time applications that can greatly enhance well-being by enabling early stress detection and mitigation, such as wearable health monitors or workplace stress management systems.

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